10 - Naive Bayes

Artificial Intelligence Lab Manual - Clean Study Edition

Learning Objectives

- Use Bayes theorem for probabilistic classification.
- Explain the conditional-independence assumption.
- Implement a basic Gaussian or text Naive Bayes classifier.

Background

Naive Bayes converts prior beliefs and feature likelihoods into posterior class scores. Although the independence assumption is often unrealistic, the method can perform well, especially for sparse text features and small datasets.

Core Concepts

Concept	Meaning in this lab
Prior P(C)	Probability of a class before observing features.
Likelihood P(X|C)	Probability of the observed features under a class.
Posterior P(C|X)	Updated class probability after observing features.
Naive assumption	Features are conditionally independent given the class.
Log probability	Used to avoid numerical underflow when multiplying many probabilities.

Guided Procedure

1. Choose either numeric classification with Gaussian likelihoods or simple text classification with word counts.

2. Estimate class priors from the training data.

3. Estimate per-feature likelihood parameters for each class.

4. Compute log-posterior scores for a new example.

5. Predict the class with the largest score and evaluate accuracy.

Reference Implementation

Use the following as a starting point. Read and modify it rather than treating it as a final answer.

```
import numpy as np

def gaussian_logpdf(x, mean, var):
    return -0.5 * np.log(2 * np.pi * var) - ((x - mean) ** 2) / (2 * var)

def class_score(x, prior, mean, var):
    return np.log(prior) + gaussian_logpdf(x, mean, var).sum()
```

Lab Exercises

Task 1. Implement Gaussian Naive Bayes from class means, variances, and priors.

Task 2. Compare your predictions with sklearn.naive_bayes.GaussianNB.

Task 3. For text, explain why zero counts cause a problem and how Laplace smoothing fixes it.

Task 4. Report precision and recall in addition to accuracy for a two-class problem.

Suggested Deliverables

- Notebook or Python file containing the completed implementation.
- Short result table or output evidence for each required comparison.
- A concise interpretation of what changed and why; do not submit output without explanation.

Review Questions

1. What is the naive assumption?

2. Why are log probabilities useful?

3. Why can Naive Bayes work even when features are not truly independent?

Completion check: You should be able to explain the algorithm or workflow in your own words, reproduce the main result, and identify at least one limitation.